



By analysing extensive X-ray data on protein interactions, the researchers unveiled intricate patterns within protein components, obscured by environmental influences. They developed a methodology to extract these patterns and unveil concealed insights, leading to the creation of a novel computer model termed Pattern Discovery and Disentanglement (PDD). This innovative PDD model has transformed pattern detection, demonstrated through numerous studies that predict patients' medical outcomes based on health records.

Technique to reform nanosensor manufacturing

Engineers at Macquarie University have unveiled a method for producing nanosensors that reduces carbon intensity and costs while improving efficiency and versatility. The



Nanoparticle sensors are smaller than human fingernails (Credit: Macquarie University)

researchers have identified a common issue wherein initial sensor fabrication often produces non-functional outcomes, necessitating adjustments, and high furnace temperatures degrade polymer-based sensors. The formation

of nanoparticles forms a weakly bonded network, creating gaps that impede electrical signal transmission, rendering the sensor non-functional. However, the introduction of a single ethanol droplet onto the sensing layer prompts surface atoms of nanoparticles to move without oven heating. This method enhances nanosensor efficiency and responsiveness, activating it within a minute, surpassing traditional 12-hour heating. The technique's success depended on precise ethanol volume determination through careful experimentation. By eliminating the heat-intensive process, this innovation opens doors for versatile nanosensor materials, revolutionising a formerly energy-intensive process into a remarkably efficient one.

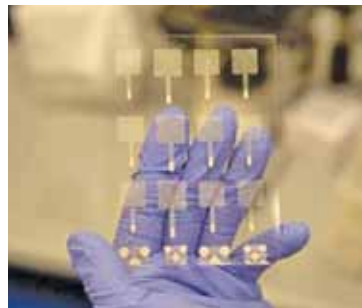
Analysing lithium movement in solid-state batteries using isotopes

A research team from Helmholtz-Zentrum Berlin, Germany and several collaborators have witnessed lithium (Li) ions' movement through the electrolyte in real time for the inaugural time. Utilising the distinct transparency of lithium ions, researchers effectively tracked their movement through the electrolyte using operando neutron radiography and in situ neutron tomography. By constructing a

lithium-sulphur battery with a lithium⁶ anode, they could differentiate mobile lithium ions from the solid electrolyte during discharge. This method highlights the non-uniform diffusion of Li and allows for faster testing of new electrolytes with the Li⁶ isotope, potentially leading to the discovery of more efficient lithium diffusion processes.

Shift towards cleaner future by hydrogen evolution reaction on gold (Au) electrodes

Researchers at Chiba University have deepened their understanding of AuNi electrocatalysts by investigating their surface structure, atomic arrangement, and hydrogen evolution reaction (HER) activity. The team assessed the HER activity of the AuNi/Au electrode in an electrochemical cell with



A set of screen-printed electrodes (Credit: Washington State University)

sulfuric acid using cyclic voltammogram (CV) and linear sweep voltammogram (LSV) measurements. X-ray photoelectron spectroscopy (XPS) and surface X-ray diffraction (SXRD) analysis revealed that the HER activity was influenced by the Au substrate's surface structure.

Ni dealloying enhanced HER activity by creating surface defects. This decrease in atomic occupancy was attributed to Ni dissolution from the surface-alloy layer, activating the HER at low-coordination Au sites adjacent to Ni. The study provides insights into the AuNi surface alloy's structural and electrochemical properties, paving the way for advanced Au-based catalysts for electrolysis and fuel cells.

A computer vision teleoperation system adaptable to various robots

Researchers at NVIDIA and UC San Diego have developed AnyTeleop, a teleoperation system driven by computer vision designed for broader application scenarios. The team created AnyTeleop, an affordable teleoperation system versatile for various robots and tasks, which translates human hand and wrist positions to robotic movements. Its compatibility spans different robotic components and environments, excelling in both close and distant scenarios. Additionally, it captures detailed human movement data, boosting robot training for autonomous tasks and outperforming specialised systems in initial tests, highlighting its transformative potential in teleoperation.